Solution:

Here there are 10 objects (letters) of which there are 3S, 3 T, 2 I and 1A and 1C

Therefore the required number of arrangements is

$$= \frac{10!}{3!2!2!1!1!}$$

$$= 50400$$

Example 7.7

If $10\,P_r = 720$ find the value of r.

$$10\,P_r = 720$$

$$10\,P_r = 10 \times 9 \times 8$$

$$10\,P_r = 10\,P_3$$

$$r = 3$$

7.3 Combinations

Combination is a selection of objects without considering the order of arrangements. For example out of three things A, B, C we have to select two things at a time. This can be selected in three different ways as follows.

AB AC BC

Here the selection of object AB and BA are one and the same. The order of arrangement is not considered in combination. Hence the number of combinations from 3 different things taken 2 at a time is 3.

This is written symbolically $3C_2 = 3$.

Now we use the formula to find combination.

The number of combination of n different things, taken r at a time is given by

$$nC_r = \frac{np_r}{r!} \quad \text{or} \quad nC_r = \frac{n!}{(n-r)!r!}$$

NOTE

(i) nC_r is also denoted by $C(n, r)$ or $\binom{n}{r}$

(ii) $nC_0 = 1$, $nC_1 = n$, $nC_n = 1$

Example 7.8

Find $10C_3$ and $8C_4$

Solution:

$$10C_3 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$$